



Free will does exist

This article argues that quantum mechanics support the existence of free will. <https://digit.in/may20-65>



Free will does not exist

This paper argues that even if the universe is indeterministic, it does not mean that free will exists. <https://digit.in/may20-66>

The radioactive material has a 50 percent chance of releasing a particle that triggers the release of the poison, over a certain period of time, say half an hour. After half an hour, opening the box will reveal if the cat is dead or alive. However, till the point you open the box, the cat is neither alive nor dead, existing in both states at the same time. Schrodinger came up with this thought experiment to illustrate the absurdity of what is known as the Copenhagen Interpretation, which was formulated by Neils Bohr and Werner Eisenberg. Think of the cat as something that can exist in one of many states, with a known probability. This is known as the wavefunction. The act of opening the box and observing the state of the cat collapses the wave function, into one of two definite values, alive or dead. According to the Copenhagen Interpretation, it was not the radioactive material, or the poison that led to the fate of the cat, but the act of opening the box!

Strangely enough, this is exactly how subatomic particles seem to behave. They exist in a cloud of probabilities of momentum and position, which are called their eigenstates. When observed, they are pinned at a particular eigenvalue. According to the Copenhagen Interpretation, the observation causes the particles to acquire the measured value. This phenomena is not to be confused with Heisenberg's uncertainty principle, which states that either the momentum or the position of a particle can be measured with accuracy. One of the experiments that seem to demonstrate the Copenhagen Interpretation is the double slit experiment. Pass a stream of photons through two slits and they form an interference pattern. Pass a photon through two slits one by one, and they still somehow form an interference pattern, interacting with other phantom photons. Pass the stream of photons through a single slit, and the interference pattern is not observed. Pass a stream of photons through two slits one by one and measure them at the slits, and

the interference pattern still does not form. The act of measuring the photons seem to decide if the interference pattern is formed or not.

Now the fact that the photons behave like both particles and waves calls into question a simple and basic tenet of classical logic - all statements are either true or false. The statement "a photon is a wave" can be either true, or false, or both. Since most of science and all of mathematics is based on classical logic, we would have to come up with an entirely new type of logic to deal with quantum phenomena. The Dancing Wu Li Masters by Gary Zukav has a description of such quantum logic. Needless to say, the Copenhagen Interpretation is considered flawed by most scientists, but they simply cannot agree on a better alternative to explain the observations and experiments.

EPR EXPERIMENT

If you are given a single shoe, and you have to predict what foot the other shoe in the pair fits, it is easy to say whether it is the right or the left shoe, by looking at the shoe in hand. In quantum physics, there are pairs of entangled particles that are like the shoes. These particles have observable states that are dependent on the other particle, even after they are separated by light years. For example if an alien on a planet in the Andromeda galaxy receives a single shoe, even they could figure out that it is one of a pair given information about human footwear. In the same way, measuring the characteristics of entangled photons are tied to the measurements of the other. For example, we can choose to measure the vertical or horizontal orientation of a single photon from a pair of entangled photons, and derive the corresponding information about the photon. Only remember that the act of observing the photon seems to pin a particular value, when they previously existed in a cloud of various probabilities. It is as if the particle knows what property the scientists are going to measure, and

somehow even manages to convey the information to its entangled twin, faster than the speed of light. Albert Einstein, Boris Podolsky, and Nathan Rosen described the situation as being incompatible with the known laws of physics, and is called the EPR Paradox. Like the Schrodinger's Cat experiment, the EPR Paradox was another attempt to question the validity of the Copenhagen Interpretation. Einstein found quantum entanglement incompatible with his own theories, and called it "spooky action at a distance". In fact, throughout their careers Einstein and Bohr took sly digs at each other. Einstein once famously said "God does not play dice", to which Neils Bohr replied, "don't tell God what to do". The takeaway in all this is that Einstein and Schrodinger were arguing that the universe was deterministic, while Bohr and Heisenberg were arguing that the universe was indeterministic.

It could be possible that all the experiments, the observers, and the particles are entangled so that the observed particles actually acquire their eigenvalues just before or at the time of the measurement. Researchers from MIT have stretched the limits of such entanglement by setting up scientific apparatus in such a way that the properties of light received from quasars 12.2 billion light years away decided which of the values of a pair of entangled photons to measure. The observer by proxy here were the quasars, which released the light even before the planet was created, the humans evolved, and the experiments were thought of. The results supported the Copenhagen Interpretation. This experiment indicates that true randomness can exist in the universe. The other possibility is that whatever mechanism the universe has for the wave function collapse, began 12.2 billion years ago when the light from the quasar was emitted, and somehow it contained the information of knowing that such an experiment would be conducted billions of years into the future, on a planet yet to be formed. 